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ENROLLMENT: MITU22BTCS0122

SUBJECT: HPCL

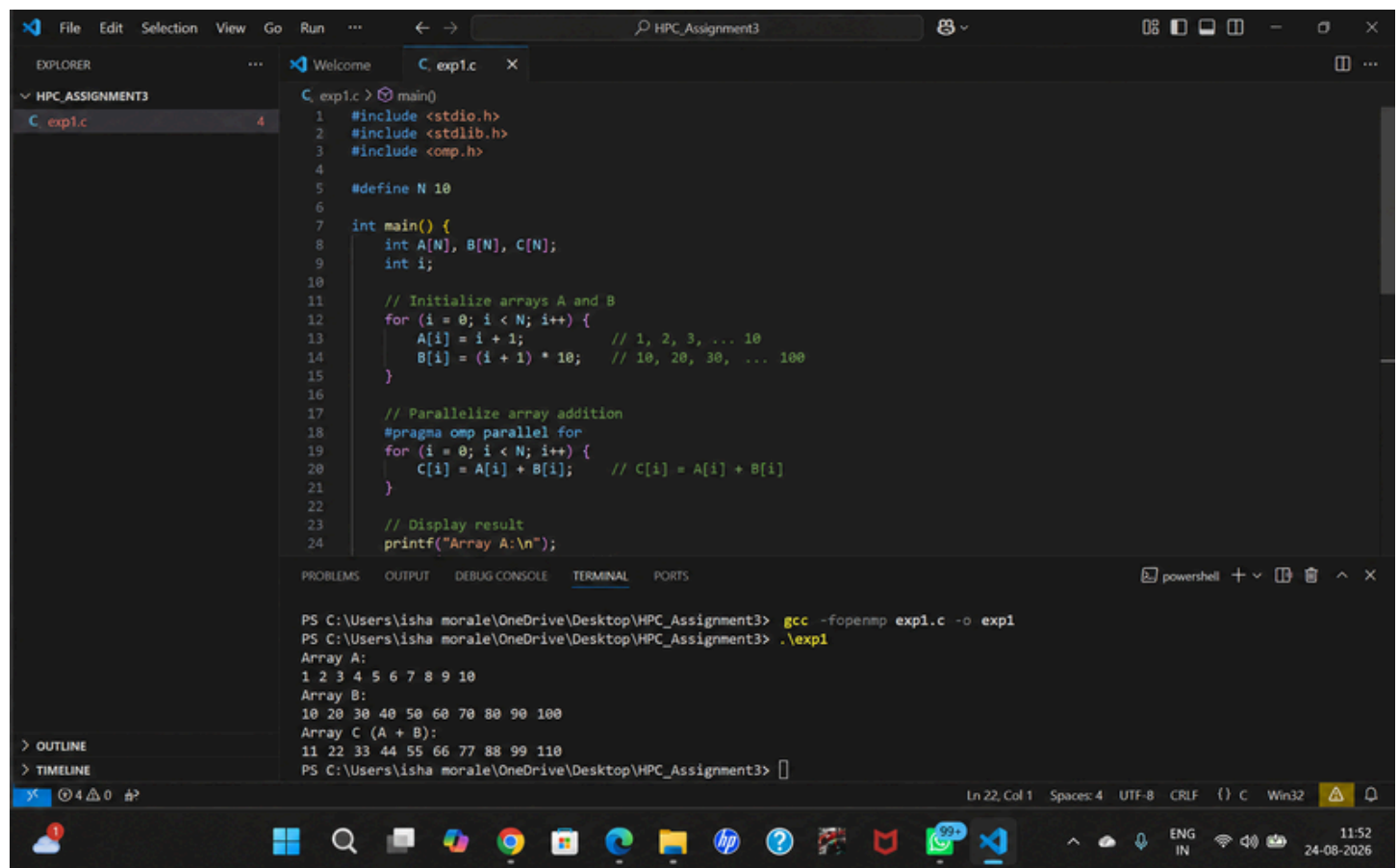
ASSIGNMENT 3

Question 1: Write a C program that adds two integer arrays A and B, each containing N elements, and stores the result in array C. Parallelize the addition using parallel for.

Solution:(CODE)

```
#include <stdio.h>#include <omp.h>#define N 10int main() {    int A[N], B[N], C[N];    for (int i = 0; i < N; i++) {        A[i] = i + 1;        B[i] = (i + 1) * 2;    }    #pragma omp parallel for    for (int i = 0; i < N; i++) {        C[i] = A[i] + B[i];    }    printf("Array C after parallel addition:\n");    for (int i = 0; i < N; i++) {        printf("%d ", C[i]);    }    return 0;}
```

Expected Output:Array C after parallel addition:3 6 9 12 15 18 21 24 27 30



The screenshot shows a Visual Studio Code window with a C program named `exp1.c` and its execution output in the terminal. The program defines `N = 10` and initializes array `A` with values 1 to 10, and array `B` with values 2 to 20. It then uses OpenMP's `#pragma omp parallel for` to parallelize the addition of `A` and `B` into array `C`. The terminal output shows the contents of arrays `A`, `B`, and `C`.

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <omp.h>
4
5 #define N 10
6
7 int main() {
8     int A[N], B[N], C[N];
9     int i;
10
11     // Initialize arrays A and B
12     for (i = 0; i < N; i++) {
13         A[i] = i + 1; // 1, 2, 3, ... 10
14         B[i] = (i + 1) * 2; // 2, 4, 6, ... 20
15     }
16
17     // Parallelize array addition
18     #pragma omp parallel for
19     for (i = 0; i < N; i++) {
20         C[i] = A[i] + B[i]; // C[i] = A[i] + B[i]
21     }
22
23     // Display result
24     printf("Array A:\n");
25     for (i = 0; i < N; i++) {
26         printf("%d ", A[i]);
27     }
28     printf("\n");
29     printf("Array B:\n");
30     for (i = 0; i < N; i++) {
31         printf("%d ", B[i]);
32     }
33     printf("\n");
34     printf("Array C (A + B):\n");
35     for (i = 0; i < N; i++) {
36         printf("%d ", C[i]);
37     }
38     printf("\n");
39     return 0;
40 }
```

Terminal Output:

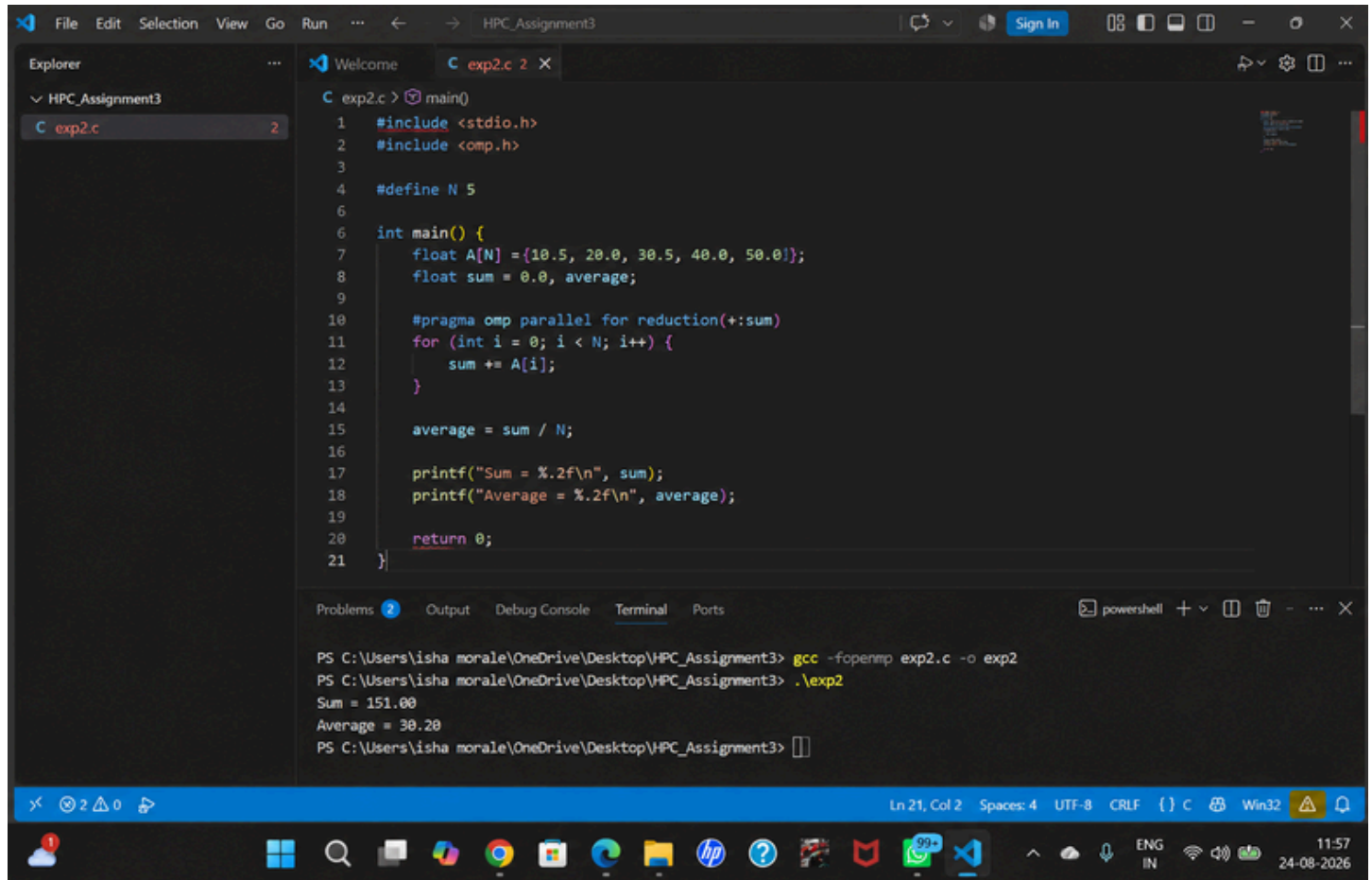
```
PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3> gcc -fopenmp exp1.c -o exp1
PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3> .\exp1
Array A:
1 2 3 4 5 6 7 8 9 10
Array B:
2 4 6 8 10 12 14 16 18 20
Array C (A + B):
3 6 9 12 15 18 21 24 27 30
```

Question 2: Write a C program that calculates the sum and average of N floating-point numbers stored in an array. Identify the shared variable and parallelize it using reduction.

Solution:(code)

```
#include <stdio.h>#include <omp.h>#define N 5int main() {    float A[N] = {10.5, 20.0, 30.5, 40.0, 50.0};    float sum = 0.0, average;    #pragma omp parallel for reduction(+:sum)    for (int i = 0; i < N; i++) {        sum += A[i];    }    average = sum / N;    printf("Sum = %.2f\n", sum);    printf("Average = %.2f\n", average);    return 0;}
```

Expected Output:Sum = 151.00Average = 30.20



```
File Edit Selection View Go Run ... HPC_Assignment3
C exp2.c 2 X
C exp2.c > main()
1 #include <stdio.h>
2 #include <omp.h>
3
4 #define N 5
5
6 int main() {
7     float A[N] = {10.5, 20.0, 30.5, 40.0, 50.0};
8     float sum = 0.0, average;
9
10    #pragma omp parallel for reduction(+:sum)
11    for (int i = 0; i < N; i++) {
12        sum += A[i];
13    }
14
15    average = sum / N;
16
17    printf("Sum = %.2f\n", sum);
18    printf("Average = %.2f\n", average);
19
20    return 0;
21 }
```

Problems 2 Output Debug Console Terminal Ports

```
PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3> gcc -fopenmp exp2.c -o exp2
PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3> .\exp2
Sum = 151.00
Average = 30.20
PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3>
```

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Question 3: Write a C program that finds the largest value in an integer array and reports its index.

Solution:(code)

```
#include <stdio.h>#include <limits.h>#include <omp.h>#define N 10int main() {    int A[N] = {12, 45, 7, 90, 23, 56, 34, 78, 11, 65};    int max = INT_MIN;    int maxIndex = -1;    #pragma omp parallel    {        int localMax = INT_MIN;        int localIndex = -1;        #pragma omp for nowait        for (int i = 0; i < N; i++) {            if (A[i] > localMax) {                localMax = A[i];                localIndex = i;            }        }        #pragma omp critical        {            if (localMax > max) {                max = localMax;                maxIndex = localIndex;            }        }    }    printf("Largest value = %d\n", max);    printf("Index of largest value = %d\n", maxIndex);    return 0;}
```

Expected Output:Largest value = 90Index of largest value = 3

File Edit Selection View Go Run ... HPC_Assignment3

Explorer

- HPC_Assignment3
 - exp1.c 4
 - exp2.c 2
 - exp3.c 3

exp3.c > main()

```
1 #include <stdio.h>
2 #include <limits.h>
3 #include <omp.h>
4
5 #define N 10
6
7 int main() {
8     int A[N] = {12, 45, 7, 90, 23, 56, 34, 78, 11, 65};
9     int max = INT_MIN;
10    int maxIndex = -1;
11
12    #pragma omp parallel
13    {
14        int localMax = INT_MIN;
15        int localIndex = -1;
16
17        #pragma omp for nowait
18        for (int i = 0; i < N; i++) {
19            if (A[i] > localMax) {
20                localMax = A[i];
21                localIndex = i;
22            }
23        }
24
25        #pragma omp critical
26        {
27            if (localMax > max) {
28                max = localMax;
29                maxIndex = localIndex;
30            }
31        }
32    }
33
34    printf("Largest value = %d\n", max);
35    printf("Index of largest value = %d\n", maxIndex);
36}
```

Problems 3 Output Debug Console Terminal Ports

PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3> gcc -fopenmp exp3.c -o exp3
PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3> .\exp3
Largest value = 90
Index of largest value = 3
PS C:\Users\isha morale\OneDrive\Desktop\HPC_Assignment3>

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